PEAS Scheme of Work: Mathematics

CLASS : SENIOR THREE

Peas

PROMOTING EQUALITY IN AFRICAN SCHOOLS

Peas

PROMOTING EQUALITY IN AFRICAN SCHOOLS

Subject: Mathematics Class: S3 Term: …ONE…. Teacher's Name: …….

Time allocation (per lesson): 3 doubles Periods per week: 6 periods

When it comes to planning your lessons use the structure: starter, I do, we do, you do, plenary. The starter is a written task that reviews previous learning. Ensure your lessons provide regular and extended opportunities for independent practice.

YPR:
Y= yes, I taught the lesson
P= I partially taught it e.g. I didn't get through all the content.
R= I taught the lesson, but I think learners would benefit from a review

S3 learning are to complete their Mathematics project in Term 1. Please indicate which weeks the learners will complete their project.

Week Subtopics	Learning outcome	Methodology	Teaching/ Learning resources	Y P R
Theme: Patterns and Algebra **Topic:** Equation of a straight line **Competency:** The learner understands and uses linear equations and their graphs				
1.1 Gradients	• The learner should be able to determine the gradient of a straight line (u, s)	• Begins the lesson by tasking leaners to study the roof of the classroom and identify the different slopes. Uses the learners' responses to introduce gradient. • Learners in pairs determine gradients from given diagrams. • Each learner determines the gradient between two given points	• Longhorn secondary mathematics book 3 page 2 – 4 • Rulers • Graph book	
1.2 Gradients to parallel and perpendicular lines	• The learner should be able appreciate and determine the gradient of parallel	• Tasks learners to identify parallel & perpendicular lines in classroom structures. Teacher uses learner' responses to draw and further explain parallels & perpendicular lines on a cartesian plane Tasks leaners in pairs to determine the	• Longhorn secondary mathematics book 3 pg 4 -7 • Graph book	

	and perpendicular lines. • (u,s)	gradients of the parallel lines and share their outcomes with the class. • Task learners in pairs to determine the gradients of perpendicular line drawn on the black board and compare their answers. • Individual learners tell with reasons whether the line AB and CD are parallel or perpendicular or neither using the points below. • (a) A(3, -1), B(6, -4), C(-4, 5), D(-2,7) • (b) A(5,8), B(7,2), C(-7,-2), D(-4,1)		
1.3 **X and Y intercepts**	• The learner should be able to determine the *x* and *y* intercepts of a given linear graph. (u,s)	• Introduce the concept of intercepts using two intersecting roads around the school compound. • Illustrates intercepts using a line drawn on the cartesian plane. • Task learners to draw determine the x and y intercepts from the drawn line. • Each leaner works on more problem sets given by the teacher • Teacher guides learners to draw the graph of 5x – 3y = 15. • Learners determine the x and y intercepts of the line. • Each individual learner draws the graph of 3x + 2y = 6 and determines the x and y intercepts.	• Longhorn secondary mathematics book 3 pg 12 - 15 • Graph books	
2.1 **Equation of the straight line** **Finding the equation of line given gradient and y intercept**	• The learner should be able to understand the relationship between a linear equation y=mx + c and its graph (k, u).	• Discusses with learners the relationship between the equation y = mx + c and the graphs • Learners work in groups to determine the gradient and y intercept of the line from the equation. • Individual learners state the equation of the line given the gradient and y intercept.	• • Longhorn secondary mathematics book 3 page16 • Graph books •	
2.2 **Equation of a line given one point and the gradient**	• The learner should be able to understand the relationship between a linear equation	• Illustrates the use of y = mx + c given m and a point (x, y) to determine the y intercept. • Task learners to substitute for m, y and x in the equation and determine the c.	• Graph books	

	y=mx + c and its graph (k, u).	• Individual learners state the equation of the line from the given values of m and c.		
2.3 Equation of the line given two points.	• The learner should be able to understand the relationship between a linear equation y=mx + c and its graph (k, u).	• Models the use of two points (x_1, y_1), (x_2, y_2) to determine the gradient of a line. • Guides learner working in groups to use any given one point to determine the value of c. • Individual learners state the equation of the line. • Learners work on more problems given by the teacher.	• Longhorn secondary mathematics book 3 pg 16 - 17	
3.1 Equation of parallel lines	• Apply the relationship of gradients of parallel and perpendicular lines to determine the equation of a straight line. (k,u,s)	• Illustrate the relationship of the gradients of two parallel lines using a ladder. • Working in groups are a given a two parallel lines A and B. A pass through two known points and B passes through one known point. They are asked to determine the gradient of line A and the equation of line B. • Learners are given the equation y = 2x + 4 and the y intercept 2 of another line L. They are tasked to determine the equation of the line L. • Teachers gives more individual tasks to learners to work out.	• Longhorn secondary mathematics book 3 pg 18. • Ladder • Long stick	
3.2 Equation of perpendicular lines	• Apply the relationship of gradients of parallel and perpendicular lines to determine the equation of a straight line. (k,u,s)	• Illustrate the relationship of the gradients of two perpendicular lines using corners of a wall. • Working in groups are a given a two perpendicular lines A and B. A passes through two known points and B passes through one known point. They are asked to determine the gradient of line A and the equation of line B. • Learners are given the equation y = 2x + 4 and the y intercept 2 of another line L. They are tasked to determine the equation of the line L. • Teachers gives more individual tasks to learners to work out. And also give the day-to-day applications of perpendicular lines.	• Longhorn secondary mathematics book 3 page 18 • Rulers • Walls of the classroom	

3.3 Activity of Integration	Teachers assess learners with activity of integration			
Theme: Geometry and Measures **Topic:** Trigonometry 1 **Competency:** The learner understands and uses the three basic trigonometric functions.				
4.1 **Trigonometric Functions** **Sine**	• The learner should be able to derive Sine from the unit circle. (u, k, s)	• Task learners to stand at the bottom of a slope. Ask the learners to measure the distance from where they are standing to the top of the slope. Ask them to also measure the angle between the ground and your line of sight to the top of the slope. Let the learner use the distance and the angle to measure the height of the slope. • Draw the unit circle on the blackboard and illustrates to the learners what Sine of an angle means. • Models to the learners how to use a calculator to determine the Sine of angles and write answers to 4 sf • Task learners to work in pairs to determine the Sine of various angles on the unit circle. And tabulate their answers. • Learners work on more problem sets in their exercise books	• Longhorn secondary mathematics learners' book 3 pg 24 – 25 • Graph paper • Pair of compasses • Ruler • Notebook	
4.2	• Use Sine in calculating lengths of sides and angles of right-angled triangles. (u, s)	• Uses worked to example to show learners how to determine the unknown sides of right-angle triangles using Sine. • Task learners to work in pairs and determine unknown sides of a triangle using Sine. • Individual learners work on more problems in their exercise books	• Longhorn secondary mathematics learners' book 3 pg 26 - 27 • Graph paper • Pair of compasses • Ruler • Notebook	
4.3 **Cosine**	• The learner should be able to derive Cosine from the unit circle. • Read and uses calculators to find values of	• Task to construct a right-angle triangle using three sticks and measure the horizontal distance and the slanting stick makes with the horizontal stick. • Draw the unit circle on the blackboard and illustrates to the learners what Cosine of an angle means.	• Longhorn secondary mathematics learners' book 3 pg 27 -28 • Graph paper • Pair of compasses • Ruler	

	trigonometric functions. (u)	• Models to the learners how to use a calculator to determine the cosines of angles and write answers to 4 significant figures. • Task learners to work in pairs to determine cosines of various angles on the unit circle. And tabulate their answers. • Learners work on more problem sets in their exercise books	• Notebook	
5.1	• Use Cosine in calculating lengths of sides and angles of right-angled triangles. • (u)	• Use worked to example to show learners how to determine the unknown sides of right-angle triangles using Cosine. • Task learners to work in pairs and determine unknown sides of a triangle using Cosine. • Individual learners work on more problems in their exercise books	• Longhorn secondary mathematics learners' book 3 pages 29 - 30 • Graph paper • Pair of compasses • Ruler • Notebook	
5.2 Tangent	• The learner should be able to derive Tangent from the unit circle. • read and uses calculators to find values of trigonometric functions. (k, u, s)	• Using the triangle learners constructed in the previous lesson as a reference, draw the unit circle on the blackboard and illustrate to the learners what Tangent of an angle means. • Model to the learners how to use a calculator to determine the Tangent of angles and write answers to 4 sf. • Task learners to work in pairs to determine Tangent of various angles on the unit circle. And tabulate their answers. • Learners work on more problem sets in their exercise books	• Longhorn secondary mathematics learners book 3 pages 30 - 31 • - Graph paper • - Pair of compasses • - Ruler • Note book	
5.3	• Use Tangent in calculating lengths of sides and angles of right-angled triangles (u)	• Use a worked example to show learners how to determine the unknown sides of right-angle triangles using Tangent. • Task learners to work in pairs and determine unknown sides of a triangle using Tangent. • Individual learners work on more problems in their exercise books	• Longhorn secondary mathematics learners' book 3 pages 31 - 33 • Graph paper • Pair of compasses • Ruler • Notebook	

6.1 **Angle of Elevation and Depression**	• Learner should be able to find the angles of elevation and depression (k, s)	• Illustrate the angle of elevation and depression using a classroom wall, a black board ruler and an observer. • Task learners to work in groups and estimate the height of a building by measuring their distance from the base of the building and the angle made by the shadow (angle of elevation). • Task learners to work in groups and determine the angles of elevation and depression from given diagrams. Present their answers to the class. • Tasks learner to sketch angles of depression and elevation from given word problems and determine distances.	• Longhorn secondary mathematics learners' book 3 pages 34 – 36. • Rulers • Protractors	
6.2 **Activity of integration**	• Assess individual learners using the activity of integration			
Theme: Data and Probability **Topic:** Data Collection or Display **Competency:** The learner collects and represents different sorts of data.				
6.3 **Introduction to Statistics** **Measures of Central Tendency** **Mean of ungrouped data**	• Understand mode, mean and median, as measures of location/ central tendency and knows how to find them and when to use them. (k, u, s)	• Ask learners to discuss the different types of data collected in their communities and guide the discussion. • Task learners to work in groups and share the types of data collected in their communities. • Ask learners to determine the average age of their group members using the formula. $\frac{\Sigma x}{n}$ • Where ∑x is the sum of their age and n is the number of members in the group. • <u>Model</u> to the learners how to determine the mean score of repeated events like tossing of die using the formula $\frac{\Sigma fx}{\Sigma f}$ Where f is the frequency. • Working in groups learners determine the mean score of the following scores in football matches of certain football team	• Longhorn Secondary Mathematics book 3 page 40 – 42	
7.1 **Median**	• Understand mode, mean and median, as measures of	• Pick five learners and ask them to stand in a straight in the order increasing heights.	• Longhorn Secondary Mathematics book 3 page 43 – 44	

<table>
<tr>
<td>Of ungrouped data</td>
<td>location/ central tendency and knows how to find them and when to use them. (k, u, s)</td>
<td>
<ul>
<li>Emphasize that the height of the person in the middle of the line is the median height.</li>
<li>Task the learners to determine the median on more sets of data using the formula $\left(\frac{n+1}{2}\right)$</li>
<li>Learners work in groups under guidance of teacher to determine the median score the above football team.</li>
<li>Individual learners solve more problems provided by the teacher.</li>
</ul>
</td>
<td></td>
<td></td>
</tr>
<tr>
<td>7.2
Mode of ungrouped data and Range</td>
<td><ul><li>Understand range as a measure of dispersion/spread and how to find it (k, u, s)</li></ul></td>
<td>
<ul>
<li>Ask the learners the items they usually buy from the school canteen and use them to explain mode of ungrouped data.</li>
<li>Task the learners to find the mode of values with odd and even numbers.</li>
<li>Draw the following table on the black board and ask learners to determine the mode.
<table>
<tr><td>Height (cm)</td><td>150</td><td>155</td><td>160</td><td>165</td><td>170</td></tr>
<tr><td>Number of Learners</td><td>4</td><td>5</td><td>9</td><td>11</td><td>10</td></tr>
</table>
</li>
<li>Explain range using scores in a maths test.</li>
<li><u>Task learners</u> to determine the range of given sets of data.</li>
<li>Each learner solves more problems on mode and range.</li>
</ul>
</td>
<td><ul><li>Longhorn Secondary Mathematics book 3 page 45 - 48</li></ul></td>
<td></td>
</tr>
<tr>
<td>7.3
Grouped Frequency distributions

Frequency Table</td>
<td><ul><li>Draw and use frequency tables for ungrouped data. (u, s)</li><li>Draw and use frequency tables for grouped data (u, s)</li></ul></td>
<td>
<ul>
<li>Using a frequency table, the explain
<ul>
<li>Class interval</li>
<li>Class mark</li>
<li>Tally</li>
<li>Frequency</li>
<li>Class boundaries</li>
<li>Cumulative frequency</li>
</ul>
</li>
<li>Task learners in pairs to form a frequency distribution table for a set of data and share with the class.</li>
<li>Each individual learner determines the class interval, class mark, cumulative frequency, modal class from the table drawn.</li>
</ul>
</td>
<td><ul><li>Longhorn Secondary Mathematics book 3 page 50 – 51</li><li>Manilla papers</li></ul></td>
<td></td>
</tr>
</table>

8.1 **Mean of grouped data**	• Calculate the mean using an assumed mean (u, s)	• Model the use of the formulae $\frac{\Sigma fx}{\Sigma f}$ and $A + \frac{fd}{\Sigma f}$ Where d is the deviation from the assumed mean A. d = x – A • Learners work in groups to determine the mean using the above formulae on given data. • Individual learner solves more problems in their exercise books	• Longhorn Secondary Mathematics book 3 page 53 - 56	
8.2 **Median of grouped data**	• Calculate the median of grouped data (u, s)	• Model the use of the formula $Median = L_m + \frac{\left(\frac{n}{2} - cf_b\right)}{f_m} \times i$ Where L_m is the lower-class boundary of median class n – Total frequency cf_b – Cumulative frequency of the class below the median class. F_m – frequency of the median class I – class interval size of the median class. • Learners work in groups to determine the median grouped data given to them and present their answers to the class. • Individual learners find the median of more data given to them by the teacher.	• Longhorn Secondary Mathematics book 3 page 52 • Manilla paper • Markers	
8.3 **Histogram**	• Draw a histogram with equal class intervals and uses it to estimate the mode (u, s)	• Demonstrate how to draw a histogram for a given frequency table and estimate the mode. • Task learners to work in pairs and draw a histogram for given frequency and estimate the mode. • Each individual learner works on more tasks given by the teacher	• Longhorn Secondary Mathematics book 3 page 57 – 58 • Graph book • Long ruler • Pencil	
9.1 **Cumulative frequency curve**	• Draw a cumulative frequency curve(ogive) and uses it to estimate the median. (u, s, v)	• Demonstrate to the learners how to draw a cumulative frequency curve (ogive) for a given frequency table and estimate the median. • Task the learners to work in pairs and draw an ogive for given frequency and estimate the median. • Each individual learner works on more tasks given by the teacher	• Longhorn Secondary Mathematics book 3 page 61 - 64 • Graph book • Long ruler • Pencil	

9.2 **Activity of integration**	• Assess learners through activity of integration.			
Theme: Geometry and Measure **Topic:** Vectors **Competency:** The learner understands the nature of vector quantities, manipulate and represent them in order to solve problems.				
9.3 **Review of Vectors** **Magnitude of vectors**	• Describe position vectors geometrically and as column vectors (k, u)	• Ask the learners the various routes they can use to come to their classroom. Use their responses to review vectors. • Explain the magnitude of a vector using directions of a given place. • Guide learners as they work in pairs to determine the magnitude of a vector using Pythagoras theorem. • Task learners to solve more problems in their exercise books.	• Longhorn Secondary Mathematics book 3 page 68 – 69 • Graph book • Long ruler • Pencil	
10.1 **Equal vectors Multiplication by a Scalar.**	• Describe position vectors geometrically and as column vectors (k, u)	• Illustrate equal vectors and how to express as column vectors. • Model multiplication of a vector by a scalar using an example. • Task learners in pairs to multiply more vectors with given scalars and share their results. • Individual learners solve more problems in their exercise books.	• Longhorn Secondary Mathematics book3 page 69 - 70	
10.2 **Position Vectors**	• Find the vector of a directed line segment when position vectors of the end points are known. (u)	• Illustrate position vectors using how learners move from home to school using a Cartesian plane. • Task learners to determine the position vector between two points on the cartesian plane. • Task learners to use the direction vectors to determine the position vector between the points. • Each individual learner solves more problem set provided by the teacher	• Longhorn Secondary Mathematics book3 page 70 – 72 • Graph book • Rulers	
10.3 **Addition of Vectors**	• Find the vector of a directed line segment when position vectors of	• Illustrate addition of vectors using short cuts taken by learners in the school compound. • Shows the learners how to add column vectors. • Task the learners to find the resultant of two or more column vectors.	• Longhorn Secondary Mathematics book3 page 73 – 74 • School compound • Measuring tape	

	the end points are known. (u)	• Individual learners work on problem sets in their exercise books.	• Ruler	
11.1 Subtraction	• Find the vector of a directed line segment when position vectors of the end points are known. (u)	• Illustrate subtraction of vectors using directions taken by learners in the school compound. • Shows the learners how to subtract column vectors. • Task learners to find the difference between two or more vectors multiplied by scalars. • Individual learners work on problem sets in their exercise books.	• Longhorn Secondary Mathematics book3 page 74 - 76	
11.2 Midpoint of a vector Proportional division of a line	• Find the position vector of the midpoint of a line segment. (u) • Use vector method to divide a line proportionately internally and externally. (s, u)	• Ask learner to use rope and determine its middle part. Build on that to explain how to determine the mid-point of line. • Gives learner two cartesian points and asks them to determine the mid-point of the line on which they lie using the formula. $\left(\frac{x_1+x_2}{2}\right), \left(\frac{y_1+y_2}{2}\right)$ • Learners divide the rope into three equal parts. • The learners then state the fraction taken by each part express it as a ratio. • Asks learners to express the vectors in ratio form. • Task individual learners to express given vectors in terms of ratios.	• Longhorn Secondary Mathematics book3 page 76 – 78 • Rope	
11.3 Parallel vectors **Collinear Points**	• Use vectors to show parallelism (s, u) • Use vector methods to show collinearity (s, u)	• Ask leaners to identify figures with parallel sides and use their responses to develop the concept of parallel vectors. • Asks learners to plot parallel points and determine their length or ratios • Task learners to find the column vectors of parallel points and express them as **AB** = K**XY** • Illustrate collinear points using trees in a straight line • Gives learners to plot three points that lie on a straight line. And find the distance between them as a ratio.	• Longhorn secondary mathematics books page 79 – 81 • Classroom walls • Graph books • Rulers	

		• Individual learners plot a parallelogram, square, rectangle etc. and show collinear point and parallel lines.		
12	**NOTE** • **Activity of integration of vectors should be in week 12** • **The mathematics projects should be complete by week 5** • **Learning aids indicated in the Scheme of Work are not limited to what has been suggest.** • **Week 12 should be used to administer Activities of Integration, marking, and preparing end of term reports.**			

Subject: MATHEMATICS Class:S3 Term: Two Teacher's Name:

Week Subtopics	Learning outcome	Methodology	Teaching/ Learning resources	Y P R
Theme: Patterns and Algebra **Topic:** Ratios and Proportions **Competency:** The learner understands ratio and proportion and use them in a range of contexts.				
1.1 **Meaning of Ratios**	• Understand and apply equivalent ratios. (u, s)	• Use the number of boys and girls in a classroom to explain ratios. • Models to the learners how to express two or more quantities in the form a: b and in the form a/b. • Learners work in groups to express the percentage fees payments in ratio form e.g., 60%, 40% and 100% and present their answers in the simplest form. • Individual learners work on more problems of expressing quantities in ratio form.	• Longhorn Secondary Mathematics Book 3 pages 84 - 85	
1.2 **Increase and decrease in a given ratio**	• Understand and apply equivalent ratios. (u, s)	• Use the increase in prices of sugar from Ugx. 4000 to Ugx. 4500 to explain increase in a given ratio. • Task learners to work in pairs to express the increase in height of a maize plant from 1.5 ft to 3 f in a given time. • Each learner is tasked to decrease Ugx. 660,000 in the ratio 11:12	• Longhorn Secondary Mathematics Book 3 pages 87 - 88	

1.3 Direct and inverse proportionality	• Understand and apply ratio, proportion and scale (k, u, s)	• Using the example of distance travelled by a car moving at a constant speed on a levelled road and the fuel consumed, explain direct proportionality. • Discuss with the learners how to express direct proportions mathematically using the proportionality symbol α • Task learner to give more example of quantities in day to day lives that vary directly. • Use the number of workers required to complete a task to explain inverse proportionality. • Shows learners how to express inverse proportions mathematically and determining the constant. • Learners work in groups to a problem on inverse proportionality. • Individual learners solve numerical problems on direct and indirect proportionality.	• Longhorn Secondary Mathematics Book 3 pages 91 - 94	
2.1 Proportional parts **Scales**	• Understand and apply ratio, proportion and scale (k, u, s)	• Share an item among some learners in the class in a particular ratio and use it to explain proportional parts. • Task learners to working groups and determine the amount of rice 3 taken by each family if they share 100 kg in the ratio 2:3:1 • Each individual learners determine the ratio of the distance travelled by John in each segment of the journey if he walks for 2 km, boards a motorcycle for the next 4 km and boards a bus for last 20 kilometres of his journey. • Use an example to demonstrate how to use proportions to work out map scales. • Learner work in pairs to Solve the problem below. On a map of scale 1:18000, the distance between a church and a school is 26.5 cm. What is the actual distance in km? • Each individual learner solves more problems involving map scales.	• Longhorn Secondary Mathematics Book 3 pages 94 – 97 • A packet of Sweets • Fruits	
2.2 Scatter diagrams	• Draw and interpret the line of best fit when looking for a	• Using a worked on a scatter graph, demonstrate how to determine the line of best fit.	• Longhorn Secondary Mathematics Book 3 pages 96 - 99	

<table>
<tr>
<td></td>
<td>relationship in bivariate data on a scatter graph (u, s)</td>
<td>• Tasks learner with points to plot the marks obtained by 8 learners in two math tests as shown below and use them to estimate the line of best fit. And also describe the correlation between the bivariate variables given to them.
<table>
<tr><td>Test 1</td><td>38</td><td>62</td><td>18</td><td>75</td><td>38</td><td>59</td><td>68</td><td>92</td><td>52</td></tr>
<tr><td>Test 2</td><td>74</td><td>44</td><td>85</td><td>19</td><td>88</td><td>69</td><td>50</td><td>33</td><td>29</td></tr>
</table>
• Each individual learner works on more data and draws scatter diagrams in their exercise books</td>
<td>• Graph books</td>
<td></td>
</tr>
<tr>
<td>2.3 Activity of integration</td>
<td colspan="3">• Assesses learners through activity of integration</td>
<td></td>
</tr>
<tr>
<td colspan="5">Theme: Geometry and Measures
Topic: Business Mathematics
Competency: The learner understands and applies business mathematics when solving problems.</td>
</tr>
<tr>
<td>3.1 Percentage Change</td>
<td>• The learner should be able to use a multiplier when calculating percentage change. (u, s)</td>
<td>• Basing on the increase in School fees from 320,000 to Ugx. 370,000, explain percentage change.
• Task learners to work in pairs and find the new price of posho if the old price of 4000 was increased by 10%.
• Learners work in pair to determine the percentage decrease in the numbers of failures in math.
• Task individual learners to solve more numerical problems on percentage change</td>
<td>• Longhorn Secondary Mathematics Book 3 pages 104 – 105
• Internet to get prevailing exchange rates</td>
<td></td>
</tr>
<tr>
<td>3.2 Compound Interest</td>
<td>• Calculate Compound interest. (u, s)</td>
<td>• Teachers uses an example of a saving of Ugx. 100000 which give a simple interest of 10% per year and all the money remains in the savings account for several years to explain the concept of compound interest.
• Working in groups, learners determine the amount of money a worker will have if he saved 100,000 and it compounded for 3 years. Using the formula of simple interest.</td>
<td>• Longhorn Secondary Mathematics Book 3 pages 106 - 108</td>
<td></td>
</tr>
</table>

		• Each individual learner uses the formula $A = P\left(1 + \frac{r}{100}\right)^n$ To determine the amount of money after 2 years with a principal of Ugx. 3,000,000 at a rate of 15% compound interest.		
3.3 Appreciation and Depreciation	• Understand and calculate Depreciation and appreciation. (u, s)	• Use an example of John who bought land at Ugx. 5,000,000 and sold it at Ugx. 9,000,000 after 3 years to explain appreciation. • Teacher also uses an example of a car bought at Ugx. 25,000,000 and sold at 20,000,000 to explain depreciation. • Task learners to use the formula of compound interest to find the new population of a town after 2 years if it was originally 150,000 and grew at a rate of 5%. • Each individual learner determines the new price of a machine which was originally priced at 2.5 m and depreciated in value at a rate of 10% after 3 years.	• Longhorn Secondary Mathematics Book 3 pages 109 - 110	
4.1 Foreign Currency Exchange	• Understand and convert local or foreign currencies. (u, s)	• Using the Us Dollar as the currency of reference, the teachers explain to learners how exchange rates work. • Uses an example and converts a given amount of Ugandan Currency to a foreign currency. • Task learners to work in pairs and Convert Ugandan Currency to US dollars, Kenya Shillings, Rwandese franc depending on the prevailing exchange rates at the time of the lesson. • Each individual learner works more problem sets given by the teacher.	• Longhorn Secondary Mathematics Book 3 pages 111 - 113	
4.2 Hire Purchase and Mortgage	• Understand and calculate hire Purchase. (u) • Describe and determines mortgage of assets. (u, s)	• Explain hire purchase to the learners using a vivid example. • Works out an example on hire purchase using Hire purchase price = Deposit + Total instalments • Learners work in groups to determine the total cost of purchasing an item using hire purchase. • Individuals determine the amount money that can be saved when an item is bought using cash than hire purchase. • Teachers discusses with the class the terms of mortgaging.	• Longhorn Secondary Mathematics Book 3 pages 114 - 118	

		• Gives learners a task of finding the amount of a loan paid by someone on mortgage terms.		
4.3 **Income Taxes**	• Calculate income tax given income Tax bands (u, v/a)	• Ask learners the to discuss the types of taxes paid in their communities. Use them to explain income tax. • Model to the learners using an example how to calculate the income tax of an individual. • Task learners to work in pairs and determine the total allowances, taxable income and the income tax given the gross income and the tax rates. • Individual learners work on more problems given by the teacher in the exercise books.	• Longhorn Secondary Mathematics Book 3 pages 119 - 122	
5.1 **Activity of Integration**	• Teacher assesses individual using an activity of integration			
Theme: Geometry and Measure **Topic:** Trigonometry 2 **Competency:** The learner understands and applies the three basics trigonometric ratios				
5.2 **Trigonometric Ratios of Angles greater than 90°**	• Determine trigonometric ratios of angles greater than 90°. (u, s)	• Ask learners to construct a circle and divide it into various parts. Use the parts to develop the concept of trigonometrical angles greater than 90° • Use a triangle within a circle to explain trigonometric ratios of angles greater than 90° and how obtain them. (quadrants) • Working in pairs, leaners draw the angles of 140° and 320° within the same diagram and use the triangles to find the ratios of sin 140, cos 140 and tan 140. • Individual learners evaluate the following. (a) Sin 147° (b) Cos 318° (c) Tan (-165°)	• Longhorn Secondary Mathematics Book 3 page 126 – 128 • Pair of compasses • Protractors	

5.3 Trigonometric Ratios of 30°, 45° and 60°	• Determine trigonometric ratios of angles: 30°, 45° and 60°. (u, s)	• Ask learner to construct an equilateral triangle and square and then discuss their properties. • Task learners to construct an equilateral triangle of side 2 cm. and bisect the triangle. Present to the class. • Individual learners find the trigonometric ratios of 30° and 60° • Using the same approach learners use a square to determine the trigonometric ratios of 45°	• Longhorn Secondary Mathematics Book 3 page • 129 – 130 • chart • Pair of compasses • Protractor	
6.1 Trigonometric Graphs	• Use the graphs of trigonometric functions to determine values of Sine, Cosine and Tangent for any angle. (u, s)	• Model the use of graphs to determine trigonometric ratios using an example. • Task learners in groups to draw a graph of the function y = cos θ and use it to determine the cosines of the angles. • Individual learners use draw the graph of the functions b = sin θ and y = tan θ for a given range and use them to determine the sines and tangents of the angles.	• Longhorn Secondary Mathematics Book 3 page • 130 – 134 • Graph Book	
6.2 The Sine rule for any triangle	• Apply Sine and Cosine rules to solve real life problems. (k, u, s)	• Use a triangle and develops the Sine rule. • Tasks the learners to use the Sine rule to find the missing sides of a triangle. • Individual learners solve more problems provided by the teachers.	• Longhorn Secondary Mathematics Book 3 page 135 - 137	
6.3 The Sine rule for any triangle	• Apply Sine and Cosine rules to solve real life problems. (k, u, s)	• Discuss more worked examples involving the Sine rule. • Learners work on pairs on a set of problems given by the teachers. • Individual learners are given further tasks to work on.	• Longhorn Secondary Mathematics Book 3 page 135 - 137	
7.1 The Cosine rule for any angle	• Apply Sine and Cosine rules to solve real life problems (k, u, s)	• Use a triangle and develops the Cosine rule. • Task the learners to use the Cosine rule to find the missing sides of a triangle. • Individual learners solve more problems provided by the teachers.	• Longhorn Secondary Mathematics Book 3 page 137 - 139	

7.2 **The Cosine rule for any angle**	• Apply Sine and Cosine rules to solve real life problems (k, u, s.	• Discuss more worked examples involving the Cosine rule. • Learners work on pairs on a set of problems given by the teachers. • Individual learners are given further tasks to work on.	• Longhorn Secondary Mathematics Book 3 page 137 - 139	
7.3 **Activity of Integration**	• The teacher assesses the learners with an activity of integration			
8.1 **Meaning of a Matrix**	• Define a matrix and state its order. (k)	• Teachers uses a stack of boxes of chalk to explain the meaning of a matrix. • Tasks in groups leaners to form matrices from the items they purchase when coming to school, state the order and present their answers to the class. • Task individual learners to write the order of a matrix and form a matrix given the order.	• Longhorn Secondary Mathematics book 3 page 142 – 144 • Boxes of chalk	
8.2 **Equality of matrices Addition and Subtraction of matrices**	• Know when matrices can be added or multiplied. (u, s)	• Explain equality of matrices and how to use them to find the unknowns. • Task learners in pairs to find unknowns of given matrices. • Model to the learner how to add and subtract given matrices using an example. • Task learners in pairs to find the sum and difference of given matrices. • Individual learners solve more problems given by the teacher.	• Longhorn Secondary Mathematics book 3 page 144 - 146	
8.3 **Multiplication of a matrix by a scalar**	• Know when matrices can be added or multiplied. (u, s)	• Model the multiplication of a matrix by a scalar using an example. • Task learners in groups to multiply given matrices by a scalar and share their answers to the class. • Individual learners are given to evaluate matrices involving a scalar.	• Longhorn Secondary Mathematics book 3 page 144 – 146 • Calculators	
9.1 **Multiplication of matrices**	• Know when matrices can be added or multiplied. (u, s)	• Use a matrix of 3 items purchased from a shop and a matrix of their prices to illustrate multiplication of matrices. Teacher emphasizes compatibility of matrices for multiplication	• Longhorn Secondary Mathematics book 3 page 144 – 146 • Calculators	

		• Task learners to work in pairs to multiply given matrices. • Individual learners are given word problems to form matrices and multiply them.		
9.2 **Identity and Zero matrices** **Determinant and Inverse of 2 x 2 Matrix**	• Find the determinant of a 2 × 2 matrix. (u, s) • Find the inverse of a 2 × 2 matrix and understands why it may not be possible to do so. (k, u, s) • Apply knowledge of matrices in solving problems from real life situation. (u, s, v/a)	• Discuss with learner the meaning of identity and zero matrices. • Task learners to write an identity matrix and zero matrix of a given order. • Use example to show learner how to find the determinant and inverse of a 2 X 2 matrix. • Learners discuss in groups the determinant and inverse of given matrices. • Individual learners are tasked to determine the inverse of matrices.	• Longhorn Secondary Mathematics book 3 page 154 – 157	
9.3 **Activity of integration**	• Teachers assesses learner through an activity of integration			
Theme: Geometry and Measures **Topic:** Matrix Transformations **Competency:** The learner understands and uses matrices to transform shapes on a coordinate grid**.**				
10.1 **Transformation represented by 2 x 2 matrices**	• Identify transformation matrices for reflection, rotation and enlargement. (k, u, s)	• Use the cartesian plane to explain the matrices of reflection, rotation and enlargement. • Task learners to plot given co-ordinates on the cartesian plane and determine the matrices of transformation. • Working individually, learners determine the images of points after reflection and rotation by given matrices	• Longhorn Secondary Mathematics book 3 page 162 – 165 • Graph paper • Ruler	
10.2 **Identifying the matrix of transformation Unit square**	• Determine the image given the object and transformation matrix,	• Model to the learners how to identify a matrix of transformation by use of unit square. • Task learners in groups to find the matrices of reflection about the x – axis and rotation of 90 about the origin.	• Longhorn Secondary Mathematics book 3 page • 166 – 168 • Graph books	

	• on a coordinate grid. (u, s) • Identify the matrix of transformation • when the object and its image are given. (u, s)	• Individual learners find other matrices of reflection in other lines and rotation in other angles about the origin. (Clockwise or anticlockwise) • Using an example of object and image discuss how to find the matrix of transformation by calculation. • Task learners to determine the matrices of transformation given the image and the object.	• Rulers	
10.3 Using inverse matrices to find the object given the image	• Determine the inverse of a transformation matrix. (u) • Use the inverse matrix to find the object when the image is given. (u, s)	• Use an example of a matrix and the image to find the object. • Gives learners a task of determining the inverse of a transformation matrix and use it find the object. • Individual learners are asked to find the object points given the transformation matrix and the image points.	• Longhorn Secondary Mathematics book 3 page • 169 – 171 • Graph books • Rulers	
11.1 Identifying the relationship between area scale factor and determinant of transformation matrix	• Identify the relationship between area scale factor and determinant of the transformation matrix. (u)	• Guide learners using a graph of object and image points to determine the area scale factor. Using the formula below. $AreaScaleFactor = \frac{areaofimage}{areaofobject}$ $Area\ Scale\ Factor = determinant\ of\ the\ matrix\ of\ transformation$ • Task learners in groups to find the area of the image given the transformation matrix and object area. • Individual learner work on more problems given by the teacher.	• Longhorn Secondary Mathematics book 3 page • 172 – 174 • Graph books • rulers	
11.2 Combined transformations	• Determines a single matrix for successive transformations. (u, v,a)	• Explain the effect of more than one transforms of an object using a relevant example. • Task learners in groups to find the single matrix after two combined transformations and describe them. • Individual learner work on more problems given by the teacher.	• Longhorn Secondary Mathematics book 3 page 174 – 177 • Graph book • Rulers	

11.3 Successive transformation involving matrices	• Determines a single matrix for successive transformations. (u, v,a)	• Explain the transformation involving two matrices one followed by the other and how to use it to find the image or object. • Give learners in a group a task of matrix K followed by L to determine the final image of two successive transformations. • Individual learners determine a single matrix of successive transformation that would map the image back to the object.	• Longhorn Secondary Mathematics book 3 page • 177 – 182 • Graph ruler	
12	**The activity of integration for matrix transformation should be given in week 12** Week 12 should be used to administer Activities of Integration, marking, and preparing end of term reports.			

Subject: Mathematics Class: S3 Term: 3 Teacher's Name:

Week Subtopics	Learning outcome	Methodology	Teaching/ Learning resources	Y P R
Theme: Patterns and Algebra **Topic:** Simultaneous equations **Competency:** The learner understands, forms and solves simultaneous equations.				
1.1 Forming Simultaneous equations	• Understands and forms simultaneous equations from word problems (u, s)	• Using the example "John has twice as many apples as Jane. Together they have 15 apples. How many apples does each person have?". Guide learners on how to form two simultaneous equations. • Learners work in groups to form simultaneous equations from a word problem. • Task individual learners to form at least two simultaneous equations from day-to-day life experiences.	• Longhorn Secondary Mathematics book 3 page 186 – 187	

1.2 Solving linear equations by substitution method	• Solves simultaneous equations using substitution. (u, s)	• Discuss with the class how to solve simultaneous equations by substituting one the unknowns. • Task learners to work in pairs and solve other sets of simultaneous equations with the help the teachers. • Individual learners solve more simultaneous equations in their exercise books.	• Longhorn Secondary Mathematics book 3 page 187 – 189	
1.3 Solving linear equations by the elimination method	• Solves simultaneous equations using elimination. (u, s)	• Discuss with the class how to solve simultaneous equations using the elimination method • Task learners to work in pairs and solve other sets of simultaneous equations with the help the teachers. • Individual learners solve more simultaneous equations in their exercise books.	• Longhorn Secondary Mathematics book 3 page 190 – 192	
2.1 Graphical method	• Draws graphs of simultaneous equations and estimates the solution. (u, s)	• Model how to use a graph and solve a set of simultaneous equations. • Learners work in groups, form co-ordinates from the lines, plot graphs and use them to solve simultaneous equations • Individual learners solve more simultaneous equations in their exercise books.	• Longhorn Secondary Mathematics book 3 page 192 – 194 • Graph book	
2.2	• Draws graphs of simultaneous equations and estimates the solution. (u, s)	• Solve more problems using graphical method of solving simultaneous equations to solidify the understanding of learners. • Task learners to discuss more problems in pairs. • Individual learners work on further problems in their exercise books	•	
2.3 Matrix method	• Uses matrices to solve simultaneous equations. (u, s, v/a)	• Discuss with the class how to solve simultaneous equations by formation of matrices • Task learners to work in pairs and solve other sets of simultaneous equations with the help the teachers. • Individual learners solve more simultaneous equations in their exercise books.	• Longhorn Secondary Mathematics book 3 page 194 - 196	
3.1 Activity of integration	• Teacher assesses learner using an activity of integration			

Theme: Data and Probability **Topic:** Probability **Competency:** The learner applies their understanding of probability to solve a wide range of problems.				
3.2 Experimental probability	• Understand the terms random, experiment, outcome, sample space, event and probability. (k,u) • Determine probabilities from experiments and real-life data. (u, s)	• Perform an experiment using a die to explain probability and discuss the outcomes with learners. • Task learners to write probabilities from events using the formula $\frac{\text{Number of favourable outcomes}}{Number\ of\ possible\ outcomes}$ • Individual use day to day situation to form find probabilities of events occurring.	• Longhorn Secondary Mathematics book 3 page 200 – 201 • A coin • A bottle tops • Dice	
3.3 Impossible and certainties	• Understand the terms random, experiment, outcome, sample space, event and probability. (k,u)	• Explain certainty and impossibilities using events like the sun rising, clouds being green. • Works out an example of picking items from the group so that the probability = number of event/ the number of sample space • Learners work in groups on more events involving picking from a group.	• Longhorn Secondary Mathematics book 3 page 201 – 203 • Bottle tops • Dice	
4.1 Theoretical probability	• Differentiate between theoretical and experimental probabilities. (k, u, s).	• Explain with an example the meaning of theoretical probabilities. • Tasks learner to give examples of theoretical probabilities and work out an example with them. • Task individual learners to find probabilities in which two events occur simultaneously	• Longhorn Secondary Mathematics book 3 page 204 – 206 • Coins and dice	
4.2 Mutually exclusive and independent events	• Identify and understand mutually exclusive and independent events (k,u)	• Tasks learner to mention any two events in day to day lives that cannot occur at the same time and also events that do not depend on each other. • Build on the above responses to explain mutually exclusive and independent events. • Tasks learner to give examples of mutually exclusive and independent events and work out an example with them.	• Longhorn Secondary Mathematics book 3 page • 206 – 210 • Coins • Dice	

		• Task individual learners to find probabilities in which two events are mutually exclusive and two events that are independent.		
4.3 **Tree Diagrams**	• Use probability trees to determine the probabilities of mutually exclusive and independent events. (u, s)	• Discuss with learners the use of a probability tree to represent events in which items are picked and replaced and those in which items are not replaced. • Learners in groups a discuss the possible outcomes of the event of picking a red or blue pen from with and without replacement using a probability tree. • Individual learners work on more tasks in their exercise books.	• Longhorn Secondary Mathematics book 3 page • 210 – 213 • Boxes • Pens	
5.1 **Probability and sets**	• Use Venn diagrams to determine probabilities. (u, s, v/a)	• Review the use of Venn diagrams to represent two sets of items. • Build on that knowledge to define regions in the Venn diagrams. • Task and guides learners to determine the probabilities from the Venn diagrams. • Individual learners find probabilities of events using a Venn diagram.	• Longhorn Secondary Mathematics book 3 page 213 – 216 • Charts	
5.2	• Use Venn diagrams to determine probabilities. (u, s, v/a)	• Use Venn diagrams to represent three sets of items. • Builds on that knowledge to define regions in the Venn diagrams. • Tasks and guides learners to determine the probabilities from the Venn diagrams. • Individual learners find probabilities of events using a Venn diagram.	• Charts	
5.3 **Activity of integration**	• Teacher assesses learners through an activity of integration			

Theme: Algebra and Patterns
Topic: Quadratic Equations
Competency: The learner understands and solves quadratic equations and relate them to the graphs.

6.1 Solving quadratic equations by factorisation	Determine the roots of: • quadratic equations using factorisation, completing the square and the formula. (k u s)	• Model how to solve a quadratic equation using the factorization method by use of an example • Task learners to work in groups to factorise a given quadratic equation using sum and product. • Working individually, each learner solves more two quadratic equations using the factorization method.	• Longhorn Secondary Mathematics book 3 page 220 - 223	
6.2 Solving quadratic equations by completing the square	• Determine the roots of: quadratic equations using factorisation, completing the square and the formula. (k u s).	• Model how to solve a quadratic equation by the method of completing the square. • Task learners to work in groups to express a quadratic equation in the form. $(ax \pm b)^2 = c$ And find the roots of the equation. • Working individually, each learner solves more two quadratic equations by completing the square.	• Longhorn Secondary Mathematics book 3 page 223 - 226	
6.3 Solving quadratic equations by the quadratic formula	• Determine the roots of: quadratic equations using factorisation, completing the square and the formula. (k u s)	• Model how to identify the values of a, b and c from a quadratic equation and solves the equation using the formula $x = \frac{-b \pm \sqrt{a^2-4ac}}{2a}$ • Task learners to solve more quadratic equation using the above formula. • Working individually, each learner solves more two quadratic equations using the formula method.	• Longhorn Secondary Mathematics book 3 page 226 - 227 • Charts • Four figure table books	
7.1 Forming quadratic equation from its roots	• Form a quadratic equation given its roots. (u s)	• Teacher works out an example on how to form a quadratic equation from its roots. • Task learners to work out form a quadratic equation from given roots. • Each individual learner solves more problems provided by the teacher.	• Longhorn Secondary Mathematics Book 3 Page 227 - 229	
7.2 Quadratic functions and their graphs	• Make a table of values for a quadratic function and draws the graph. (k u s)	• Teachers models how make a table of values for a quadratic function and draw the graph. • Tasks learner to work in pairs and draw graphs for quadratic function. • Each learner is tasked with plotting two graphs of two other function in his or her graph books	• Longhorn Secondary Mathematics book 3 page 230 – 234 • Graph book	

7.3	• Link the solutions of a quadratic equation with its graphical representation. (k u s)	• Use an example to show learners how to link the solution of a quadratic equation with its graphical representation. • Task learners to work in pairs to solve the quadratic equation by the graphical method. • Individual learner work on more problem sets given by the teacher.	• Longhorn Secondary Mathematics book 3 page • Graph books	
8.1 Simultaneous equations: One linear, one quadratic	• Solve simultaneous equations involving one quadratic equation and a linear equation. (u s)	• Model how to plot a line and curve on the same axes and discusses with the learners how to find the roots of the equation. • Learners pair up and solve two simultaneous equations with one of them being a quadratic. • Individual learners solve more problems provided by the teacher.	• Longhorn Secondary Mathematics book 3 page 235 – 238 • Graph book	
8.2 Activity of integration	• Assess the learners using an activity of integration			
Theme: Geometry and Measures **Topic:** Circle Properties **Competency:** Learners should be able to understand and use circle properties to solve problems.				
8.3 Definition of circles Length of an arc	• Identify arc, chord, sector and segment	• Teacher uses the concept of road roundabout to explain the concept of a circles. • Models to the learner how to get the length of an arc. • Tasks the learners to work on more problems in pairs. • Gives individual learner a task of find the length of a give arc.	Longhorn Secondary Mathematics book 3 page 239 – 242 - Pair of compasses - Protractor	
9.1 Area of a sector Area of a triangle using trigonometry	• Relate angles made by an • arc at the circumference • and centre. (u) • Calculate the area of sectors and segments. (u, s, v/a)	• A teacher cuts a chapatti into pieces through the centre to explain the area taken by a sector. • Discuss with the learners the formula for getting the area of the circle and use it to get the fraction of a sector. • Task learners to work in pairs and find the area of a sector. • Individual learners work on more problem sets given by the teacher.	Longhorn Secondary Mathematics book 3 page 242 -245 - Pair of compasses - Ruler	

		• Model to the learners how to determine the area of a triangle using trigonometry • Task learners to determine the area of more triangles using trigonometry given their sides.		
9.2 Chord properties of Circles Area pf a segment	• Determine the Tangent, chord and angle properties of the circle. (u) • Find the length of the common chord for two intersecting circles. (u s)	• Discuss the chord and its properties with the learners. • Works out an example on determining the length of the chord, distance from the centre and the radius of the circle. • Task learners to work in pairs on more problems given by the teacher. • Individual learners solve more problems given by the teacher.	Longhorn Secondary Mathematics book 3 page 246 – 250 - Pair of compasses - Ruler	
9.3 Angle properties of circle	• Determine the Tangent, chord and angle properties of the circle. (u) • Find the length of the common chord for two intersecting circles. (u s)	• Discuss the angle properties of a angles at the centre and at the circumference The teachers work out an example on determining the angle at the centre made by the same chord at the circumference • Task learners to work in pairs on more problems given by the teacher. • Individual learners solve more problems given by the teacher.	Longhorn Secondary Mathematics book 3 page 250 – 252 - Pair of compasses - Ruler - Protractor	
10.1 Discovering the angle properties of a cyclic quadrilateral	• Determine the uses of: Properties of a cyclic & quadrilateral. (u s)	• Discuss the properties of cyclic quadrilateral with the learners. • Works out an example on determining the angles of the opposite sides of a quadrilateral. • Task learners to work in pairs on more problems given by the teacher. • Individual learners solve more problems given by the teacher.	Longhorn Secondary Mathematics book 3 page 252 – 254 - Protractor	
10.2 Tangent properties of a circle	• Determine the uses of: Properties of a cyclic & quadrilateral. (u s)	• Asks learners to study a bicycle chain mechanism and discover the tangential points. • Discuss properties of Tangent to the radius and uses it to find the distance and the angles formed. • Task learners to work in pairs and determine the distance to a Tangent from a given point. • Learners work out the angles formed by a Tangent.	Longhorn Secondary Mathematics book 3 page 255 – 260 - A bicycle - Ruler	

10.3	• Determine the uses of: Properties of a cyclic & quadrilateral. (u s)	• Illustrate to the learners that two tangents drawn from a common point are equal. • Works out an example to determine the angles formed. • Task learners to work in pairs and determine the distance to a Tangent from a given point. • Learners work out the angles formed by a Tangent. • Teacher emphasizes only the alternate segment theory and uses it to find the angles made by the Tangent. • Learner is given more problem set to work on.	- Pair of compasses - Ruler - Protractor	
11-12	**Activity of integration for Circles** Week 11 and 12 should be used to administer Activities of Integration, marking, and preparing end of term reports. In addition, this time should be used for end of year assessment.			